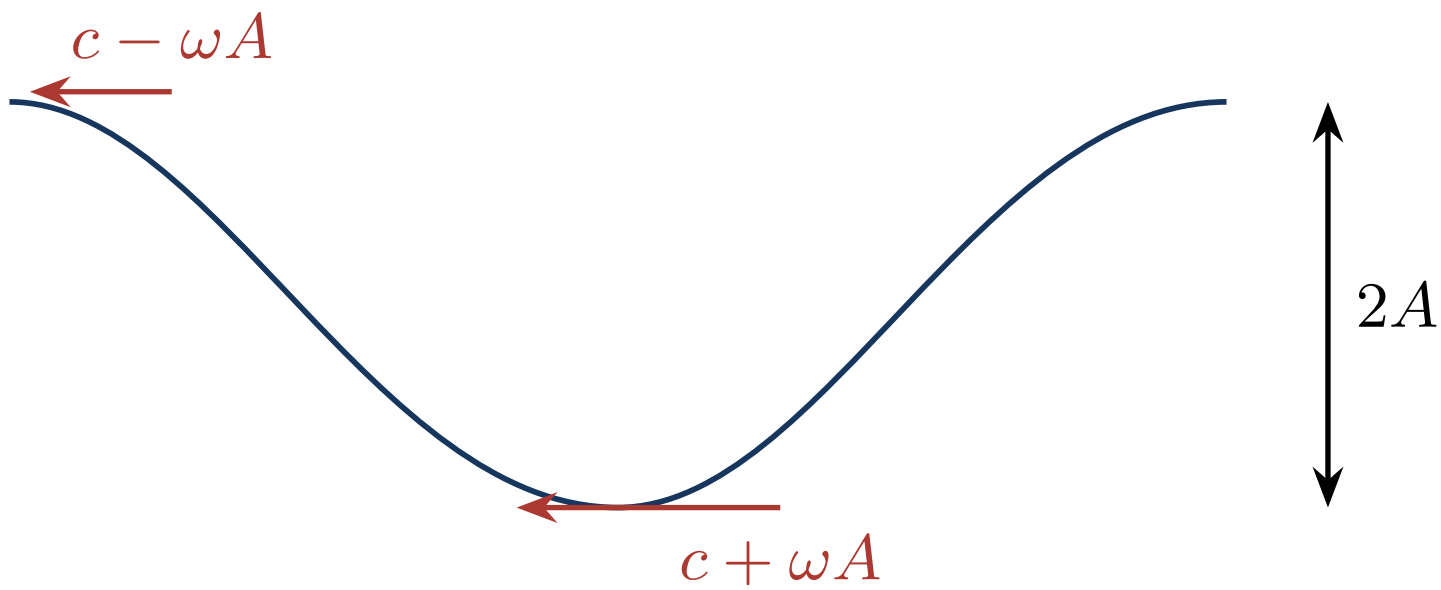
自由面两处压强同为 p_0



$$\frac{1}{2}[(c+\omega A)^2-(c-\omega A)^2] = 2gA$$

$$\text{故 } c\omega = g$$